

"Mathematics and Genius" by Nathan A. Court, in *Mathematics in Fun and in Earnest*, Dover Publications, Inc., 1958, pp.75-83.

2 • Mathematics and Genius

A • The "Heroic" and the "Objective" Interpretations of History Once upon a time, many, many years ago, so the story goes, a beautiful stallion was brought to the royal court and presented to the king. The stallion was very wild. The king was warned that no man had ever managed to mount the fiery beast. The heir apparent who happened to witness the presentation ceremony of this unusual gift, jumped upon the back of the spirited horse, and before anybody had time to realize what was happening, the young prince was already way out of sight. The king's anxiety for the safety of his beloved son was very great. After a certain lapse of time, the young man reappeared, safe and sane, on the back of the subdued, tame animal. The proud and loving father was so elated that he exclaimed in exaltation: "My son, find for yourself another kingdom. Mine is too small for you." These accidental words of the king took deep root in the sensitive soul of the young prince. History knows this young man under the name of Alexander the Great (356-323 B. C.), the famous conqueror of the ancient world.

I read this story in my school-text on ancient history, a

fine book, full of names and dates. Every historical event had its precise moment of occurrence recorded. You were told exactly by what king, or general, or by what great leader any given event was brought about. For the sake of brevity let us refer to this way of conceiving historical events as the "heroic view" of history.

This heroic interpretation of history is very attractive, because of its simplicity and its definiteness. All the whys and wherefores are readily answered by the names of the great men who made the history of the nation, or of the race. However, this heroic view has an obvious weakness: it makes history whimsical, capricious, and accidental, to the point of triviality. Suppose that our stallion of a moment ago, in its frantic effort to rid itself of its unsuccessful and unlucky tamers, had broken a leg, or two. King Philip would have been deprived of the occasion to utter those fateful words of his, and his son Alexander would have lived out his life as an obscure and inconsequential ruler of the little kingdom of Macedonia.

According to a much repeated saying, of undetermined origin, "God made George Washington childless, so he could become the father of his country." Thus, if it were not for some physiological peculiarity or deficiency of Martha Washington (or was it of George himself?) this country would have remained a British colony, even unto this very day and generation.

During the nineteenth century various writers, like the Englishman Henry Thomas Buckle (1821-1862), the Frenchman Hippolyte Taine (1828-1893), best known in the English speaking world for his history of English literature, the German Karl Marx (1818-1883) have advanced the view that human history is not made by individuals, but is dominated by objective factors, like climate, geographic environment, natural resources, economic and social conditions, etc. This objective interpretation of history has since gained a great deal of ground. A forceful presentation of this conception may be found in the presidential address delivered before the American Historical Association by Edward P. Cheyney (1861-1947), under the title "Law in History" in which the following two passages occur: "History, the great course of human affairs, has not been the result of voluntary action on the part of individuals or groups of individuals, much less of chance, but has been subject to Law." "Men have on the whole played the part assigned to them: they have not written the play. Powerful rulers and gifted leaders have seemed to choose their policies and carry them out,

but their choice and success with which they have been able to impose their will upon their time have alike depended on condition over which they have had no control".

The heroic and the "objective" interpretation of history are obviously poles apart. Which of them is right? General human history is so many-sided, so complex, that it is easy enough to emphasize one element or another of its vast contents and arrive at conclusions which are contradictory, and still have each a good deal of truth in them. We may try to simplify the problem, as we often do in mathematics, reduce the number of variables, and examine a few of them at a time.

Our objective may perhaps be achieved more readily if we examine the history of a restricted, particular domain, say, that of mathematics.

B • Are Inventions Inevitable? We are accustomed to pronounce with respect and admiration, not to say with reverence and awe, names like Euclid, Archimedes, Descartes, Newton, Leibniz, Lagrange, Gauss, Poncelet, Klein, Poincaré, and many others. We know the books those men have written, the theorems which bear their names. In our own time we know by name men who live in our midst and some of whom we know personally, men who lend luster and glory to our generation, men who give us courage and inspiration. Through the study, direct and indirect, of the works of these eminent scholars we know what they have contributed to the growth and advancement of mathematical science. There hardly can be a more forceful confirmation of the importance of the individual in history, of the heroic interpretation of history, if you will. Nevertheless, there is another side to this medal.

On December 21, 1797, in Paris, the great mathematicians Laplace and Lagrange were both present at a brilliant social gathering which included a great many celebrities. Among the guests was also a victorious young general whose star was ascending rapidly, and who happened to be a former student of Laplace. In the course of the evening the general, while talking to the two world famed scholars, entertained them with some unusual and curious solutions of well known problems of elementary geometry, but solutions with which neither of his two eminent listeners were familiar. Laplace, a bit peeved, finally said to his erstwhile pupil, "General, we expect everything of you, except lessons in geometry." The name of the young general was Napoleon Bonaparte. Napoleon had learned about those strange con-

structions during his famous campaigns in Italy, whence he had just returned. While there, he met Lorenzo Mascheroni, a professor at the University of Pavia, who that very year, 1797, published a book *Geometria del Compasso* in which the author showed that all the constructions that can be carried out with ruler and compass can also be carried out with compass alone, a very astonishing result, indeed. Had Mascheroni died in infancy, would science have been deprived forever of those Mascheronian constructions? One may think the question preposterous, for such a hypothetical query admits of no answer, one way or the other. Curiously enough, in the present case the question can be answered, in a very definite way. A century and a quarter before the publication of Mascheroni's book a Danish mathematician Georg Mohr published in Amsterdam a book in two languages, one in Danish and the other in Dutch, simultaneously, in which he gives Mascheroni's main result, as well as the solutions of a good many of the problems solved later by the Italian scholar. Mohr's book passed entirely unnoticed by his contemporaries. It came to light in the present century by accident. In the preface to his book Mascheroni states explicitly that he knows of no previous work along the same lines as his book, and there is not the slightest reason to doubt his word.

The story emphasizes the fact that so many mathematical discoveries, great and small, have been made independently by more than one scholar. This multiplicity of claims to the discovery of one and the same thing is probably the most outstanding fact in the history of mathematics.

The dispute as to whether Newton or Leibniz invented the calculus is well known.¹⁰ The French claim, with a good deal of justice, that Fermat anticipated both of them. It is only Fermat's strange and persistent aversion to the pen that deprived him of the credit as inventor of that powerful mathematical tool.

A similar story may be related about the epoch-making discovery of analytic geometry. There is as much reason to refer to this discovery as "Fermatian" as there is to call it "Cartesian." Carl B. Boyer in the preface to his *History of Analytic Geometry*¹¹ says: "Had Descartes not lived, mathematical history probably would have been much the same, by virtue of Fermat's simultaneous discovery (of analytic geometry)."

The geometric interpretation of complex numbers was discovered independently and almost simultaneously by four different men, at the beginning of the nineteenth century. An

instrument for drawing a straight line without the use of a ruler, known as the "cell of Peaucellier" (1832-1913), was also invented by a young student Lipkin of St. Petersburg (Leningrad).¹²

Even whole theories have grown up, the paternity of which nobody can claim with justice. A good and simple example of this kind is offered by the theory of inversion. This theory came into being early in the nineteenth century, and from so many different quarters that it is impossible to associate any particular name with it. The only thing that can be said about it is that, like Topsy, it "just grewed."

The multiplicity of claims to the same discovery is so common that not only have we stopped to be surprised by it, but we have grown accustomed to expect it. Better to be able to protect the priority right of contributors, most of the editors of mathematical journals add to each article they publish, the date when that paper was received in the editorial office.

What was said about mathematics may be repeated with equal force about astronomy, physics, chemistry, mechanics, in fact about any science, pure or applied. Two industrious sociologists compiled a list of inventions, each of which has more than one claimant to its paternity. The list contains 148 entries and is far from being exhaustive. Armed with their incredible, but correct list, the two authors fire, point blank, an amazing question at their readers, namely: "Are inventions inevitable?"¹³

C • Genius and Environment We are prone to think that the essence of genius is freedom. Does not genius invent or create what he will? On closer examination, however, it is seen that this conception of genius is an exaggeration. What a genius may accomplish depends upon circumstances which can be controlled by no individual. The invention of the creative individual is necessarily an extension of the knowledge of his time, or is something that satisfies the needs of his contemporaries. These characteristics have to be incorporated in the invention, if the genius is to be recognized as such. If a self-taught scholar from somewhere in the hinterland would send to the editor of a journal or to the Academy of Science a manuscript which in substance would amount to the discovery, say, of non-Euclidean geometry, or of the sextant, not much fuss would be made about the author, even if his honesty would not be called into question. And such things happen, on various levels of achievement. About the middle of the nineteenth century the Academy of

St. Petersburg was offered by a teacher in some rural elementary school a crude exposition of the basic ideas of the calculus.

On the other hand, what genius can accomplish depends upon what others have done before. Newton realized that if he had seen farther than others, it is because he was "standing on the shoulders of giants." "Perhaps nowhere does one find a better example of the value of historical knowledge for mathematicians than in the case of Fermat, for it is safe to say that, had he not been intimately acquainted with the geometry of Apollonius and Viète, he would not have invented analytic geometry."¹⁴ On the other hand, as great a genius as Archimedes could not invent analytic geometry, for the algebraic knowledge necessary for such an achievement was not available in his time.

The relation between the genius and the culture he is born into is expressed by A. L. Kroeber in the following way:¹⁵ "Knowing the civilization of a land and of an age, we can then substantially affirm that its distinctive discoveries, in this or that field of activity, were not directly contingent upon the personality of the actual inventors that graced that period, but would have been made without them; and that, conversely, had the great illuminating minds of other centuries been born in the civilization referred to instead of their own, its first achievements would have fallen to their lot. Ericsson or Galvani, eight thousand years ago, would have polished or bored the first stone; and in turn, the hand and mind whose operations set in inception the neolithic age of human culture, if held in its infancy in unchanged catalepsy from that time until today, would now be devising wireless telephones and nitrogen extracts," or (let us add) nuclear weapons and interstellar ships, a generation or two later.

The dependence of the individual, whatever his natural endowments, upon the time and civilization he happens to live in, becomes quite obvious, once attention is called to this phenomenon. We are not a bit surprised to see that the French children are so very partial to the French language, and that the Chinese children, not to be outdone by the French, speak as unanimously the Chinese tongue. The same may be said, in a broader sense, about arts and crafts, music, or any other component element of culture. On a larger scale, analogous remarks may be made about those parts of culture which have become common to a considerable part of mankind, like the sciences, and mathematics in particular.

These observations may help us to comprehend the reasons for the multiplicity of claims for the same discovery. The anthropologist Leslie A. White puts it this way:¹⁶ "In the body of mathematical culture there is action and reaction among the various elements. Concept reacts upon concept: ideas mix, fuse, form new syntheses. When this process of interaction and development reaches a certain point, new syntheses are formed of themselves. These are, to be sure, real events and have their location in time and space. The places are, of course, the brains of men. Since the cultural process has been going on rather uniformly over a wide area of population, the new synthesis takes place in a number of brains at once."

Tobias Dantzig (1884-1956) in his admirable book, *Number—The Language of Science*,¹⁷ says the same thing, with a different emphasis: "It seems that the accumulated experience of the race at times reaches a stage when an outlet is imperative and it is merely a matter of chance whether it will fall to the lot of a single man, two men, or a throng of men to gather the rich harvest."

D • Genius and the "Instinct of Workmanship" Granting that objective conditions determine the kind of discoveries that can be made at any given period of history on the one hand, and that on the other hand such inventions are "inevitable," such forward steps do not take place automatically. Each particular advance requires an effort, and often a very strenuous one, on the part of the gifted individual, the "genius" who brings it about. What impulse does the individual respond to, when he makes the requisite effort?

Mathematics, like any other science, in its early stages developed empirically for practical, utilitarian purposes. The demand for its services never cease through the ages, although the extent and the pressure may vary widely from one period to another, and mathematical inventiveness may vary accordingly. This is quite clear, for instance, in the case of the rapid strides made by mathematics during the brilliant seventeenth century.¹⁸ The mathematician, like any other scientist, is not unmindful of the needs of his time and is not indifferent to the acclaim that would be his if he supplied the answer to a pressing question of his day.

There is, however, another phase of the situation to be considered. After a sufficient amount of mathematical knowledge has been accumulated, the cultivation of this domain of learning may become an interest in itself. Those versed in its secrets and adept in manipulating them may

find it attractive to strive for new results just to satisfy what Thorstein Veblen (1857-1929) called the "instinct of work-manship."¹⁹ The only extraneous element in the case may be the wish to gain the approval of the restricted audience of likeminded people or perhaps to confound some rivals. Outside of that the reward that may accrue to the mathematician for his efforts is to live through the pains of creation and to experience the exhilarating joy of discovery. His is a labor of love. He considers himself amply repaid if he feels that he added, be it ever so little, to the luster of the brightest jewel in the intellectual crown of mankind—The Science of Mathematics.

E • Mathematics—the Patrimony of the Race Our discussion has thus led us to ascribe less importance to the role of the individual in the development of mathematics and to give more credit for the creation of this magnificent edifice to the human race as a whole. To be sure, it is always through the gifted individuals that the progress takes place. But no individual is indispensable in this task of furthering mathematical knowledge. The human race produces enough ability of a high degree to make the progress independent of any individual. Albert Einstein said in a press interview: "Individual worship, as I look at it, is always something unjustified. To be sure, nature does distribute her gifts in rich variety among her children. But of those richly gifted ones there are, thank God, many, and I am firmly convinced that most of them lead a quiet unobtrusive existence."

Mathematics is the patrimony of the human race. It is the result of slow and patient labor of countless generations over a period of a great many centuries. Various practical callings have contributed towards this accumulation of mathematical knowledge and have furthered its development in the early and difficult stages. Modern technology provides such stimulation at an ever accelerating pace. The effort which has been expended in erecting the stately and imposing structure which we call mathematics is enormous. But mathematics has repaid the race for this effort. The practical value of mathematics cannot be overemphasized. To those privileged to appreciate the intellectual greatness of mathematics, the contemplation of this grandeur is an endless source of pure joy. The esthetic appeal of mathematics has found its enthusiastic and eloquent exponents. It would be proper to mention here another phase of the merit and value of mathematics to mankind.

The superiority of the human race over all the creatures inhabiting the earth, the reason that mankind is the master of this globe is due primarily to the fact that the experience of each generation does not die with that generation, but is transmitted to the next. This transmission of accumulated experience from generation to generation is the real power of the race, its greatest asset, its most powerful weapon in the conquest of nature, its surest tool in the accumulation of intellectual treasures. Nowhere is this more manifest than in mathematics. The cumulative character of mathematics is really astonishing. There is little in mathematics that ever becomes invalid, and nothing ever gets old. We may have all sorts of non-Euclidean geometries, non-Archimedian geometries, n -dimensional geometries, but all this makes the venerable elements of Euclid neither invalid nor obsolete. They remain, graceful and solid, an object of studies as much as ever, all in their own right. This cumulative process, this constant enlargement and perfectibility of mathematics, is the most precious of its characters, for it has given to mankind the idea of progress, with a clearness and distinctness that nothing else can equal, let alone surpass.

C. J. Keyser in his book *Humanism and Science*²⁰ goes a step farther and points out that the idea of progress suggested by science, and particularly by mathematics, has reflected upon the race itself. It has given mankind the idea that human nature in its turn may be perfected, that with the growth of knowledge and improved living conditions the human race will keep on rising to greater and greater heights on the road toward civilization. Mathematics has given the human race not only the technical tools to bend nature to its uses, not only a great and unequalled storehouse of intellectual beauty and enjoyment, but it also has given mankind a faith in itself and its destinies, hope and courage to carry on this unceasing struggle for a better, more noble, and more beautiful life.

FOOTNOTES

¹ Cf. Chapter VII, Section 1b.

² *Handbook of South American Indians*, James H. Steward, editor, Vol. 5 (Smithsonian Institution, Washington, D. C., 1929) p. 614.

³ For instance, *What Happened in History*, V. Gordon Childe, (Pelican Books, A 108, London and Baltimore).

⁴ Cf. Chapter I, Section 1b.

⁵ Ibid.

⁶ See Chapter III, Section 3c.

⁷ Cf. Chapter III, Section 3h; Chapter VI, Section 2e.

⁸ See Chapter V, Section 2b.

⁹ "Law in History", *American Historical Review*, Edward P. Cheyney, Vol. 29 (1923-1924), pp. 231-248.

¹⁰ Cf. Chapter V, Section 1c.

¹¹ *History of Analytic Geometry*, Carl B. Boyer (Scripta Mathematica Studies, New York, 1956).

¹² *Outline of the History of Mathematics*, R. C. Archibald, (Mathematical Association of America, 1949), p. 99, note 280.

¹³ "Are Inventions Inevitable?" William F. Ogburn and Dorothy Thomas, *Political Science Quarterly*, Vol. 37 (1922), p. 83.

¹⁴ *History of Analytic Geometry*, op. cit.

¹⁵ "The Superorganic", A. L. Kroeber, *The American Anthropologist*, Vol. 19 (1917), p. 201.

¹⁶ "The Locus of Mathematical Reality. An Anthropological Footnote", Leslie A. White, *Philosophy of Science*, Vol. 14, No. 4 (October, 1947), p. 298.

¹⁷ *Number—The Language of Science*, Tobias Dantzig (First edition, New York, 1930), pp. 195-196.

¹⁸ Cf. Chapter II, Section 1c.

¹⁹ *The Instinct of Workmanship*, Thorstein Veblen (New York, 1914).

²⁰ *Humanism and Science*, C. J. Keyser (Columbia University Press, New York, 1931).